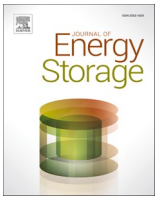
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Research Papers



Experimental study on photovoltaic panel integrated with Polyethylene Glycol 1500 phase change material

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ABSTRACT

The heat generated during solar photovoltaic (PV) operation increases the PV cells temperature and reduces electric power generation. Several cooling strategies have been used to remove the heat generated during the process. One of them is the addition of a layer of phase change materials on the backside of PV panel, which can control PV temperature rise and result in increased electrical conversion efficiency and power output. In this paper, the use of novel Polyethylene Glycol 1500 (PEG 1500) as phase change material (PCM) has been explored. Also, the effect of metal matrix insertion into phase change material integrated with PV has been presented. A comparative study has been done for PV/PCM with and without metal matrices. The Polyethylene Glycol 1500 has a melting temperature in the range of 50 °C–56 °C. PV panel's thermal and electrical performance in outdoor conditions is experimentally investigated. The results indicate that use of PEG 1500 as PCM decreases PV panel temperature by 10.59 % and 14.08 % and enhances electrical efficiency by 5.84 % and 8.87 % for PV/PCM and PV/PCM metal matrix systems, respectively. However, the power output of 40 W panel increases by 2 W and 3.5 W for PV/PCM and PV/PCM metal matrix systems, respectively, when the solar intensity is at its peak.

1. Introduction

The energy requirement in the 21st century is the most relevant concern among the investigators who work on an energy-efficient solution and its supervision. In 2018, the world energy consumption reached around 22,315 TWh, which was 4 % higher than in 2017. From 1974 to 2017, an average growth rate of 3.3 %/year was observed in energy consumption [1]. The huge potential in renewable energy sources nowadays overcomes a load of energy production from fossil fuels by reducing CO_2 emissions. India has pledged to provide 40 % non-fossil electricity power by 2030 with a 30–35 % reduction in emission intensity. Fig. 1 shows the expected total renewable energy installed capacity of 275 GW by 2027. Out of 275 GW of renewable energy, solar energy has a share of 175 GW [2]. This indicates that solar energy has enormous potential for electrical energy generation in the coming decades.

Solar energy is the most favorable energy source among the other available renewable energy sources for power generation. Solar energy is known for being clean, freely available in all parts of the world and most importantly, it is environment friendly. Solar energy can be converted to electrical energy through a photovoltaic system and a concentrated solar photovoltaic system. However, electric power generation through a photovoltaic system is a promising technology. But the major disadvantage is that electrical conversion efficiency decreases with the increase in PV cell temperature. In summer conditions, PV panel temperature may reach 65–85 °C and electrical conversion efficiency reduced by 0.45–0.65 %/°C [3]. Hence, to overcome this major limitation, many researchers have studied the different active and passive cooling techniques to keep PV cell temperature as low as possible [4–7]. An active cooling system uses a fan, blower or pump, whereas a passive cooling system does not use any external power to run the system. Passive cooling offers an advantage over active cooling since it is less expensive, easier to install, and consumes no energy. Active cooling systems have the capacity and capability to control the cell temperature under any climatic circumstances, as long as the system's economics is not taken into account. The techno-economic feasibility and performance evaluation of active and passive cooling based systems are reported by Nizetic et al. [8,9].

One of the most effective passive cooling methods for a standalone PV system is the use of phase change materials (PCMs) [10]. PCM stores and releases a large amount of heat during the phase change process. The phase change materials have numerous applications, such as thermal comfort for buildings, electronics cooling and refrigeration systems

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Nomenclature

A	area (m^2)
G	solar irradiance (W/m^2)
I_m	maximum panel current (A)
Isc	short-circuit current (A)
P_m	maximum power output (W)
PCM	phase change material
PEG	Polyethylene Glycol
PV	photovoltaic
RTD	resistance temperature detector
SHP	small hydropower
STC	standard test condition
Ta	ambient temperature (°C)
Tp	temperature of panel with PCM cooling (°C)
Tpm	temperature of panel with PCM + Matrix cooling (°C)
Tref	reference panel temperature (°C)
U	uncertainty parameter
V_m	maximum panel voltage (V)
Voc	open-circuit voltage (V)
WTE	waste to energy
η_{pv}	PV panel efficiency (%)
β_{ref}	temperature coefficient (1/K)
η_{ref}	reference efficiency (%)

Subscripts

m	max
p	PCM cooled
pm	PCM matrix cooled
pv	photovoltaic
ref	reference

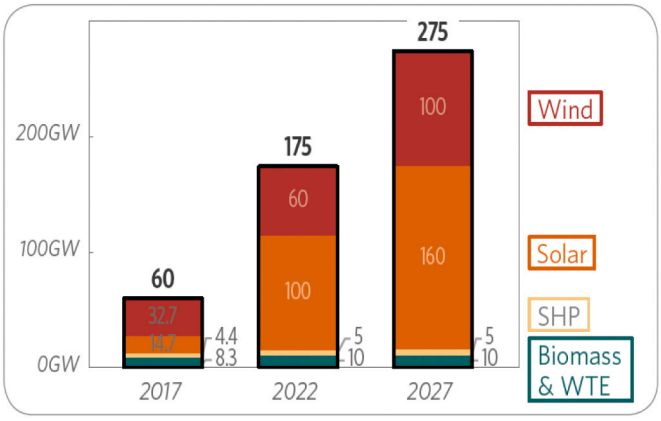


Fig. 1. Share of renewable energy in electrical energy generation [2].

[11–13], cold and heat storage [14], and thermal management of batteries and PV systems [15,16].

2. PV thermal management using phase change materials

The experimental and numerical studies that have been carried out using phase change materials in PV panel thermal management in different ways are described in subsequent sections.

2.1. PV + PCM rectangular enclosure only

PV panel with PCM rectangular enclosure is one of the popular methods of incorporating PCM on the backside of PV panel. It uniformly extracts heat from the PV panel. Stropnik et al. [17] have shown how to improve PV panel efficiency by integrating a PCM rectangular enclosure. The RT28HC as PCM was tested in Slovenia climatic conditions experimentally and validated with a numerical model. The result shows a 9.2 % increase in electrical power production compared to the reference panel. Hasan et al. [18] explored the use of paraffin as PCM in the United Arab Emirates' hot climatic conditions to assess the year-round energy-saving performance of PV panels. A conjugate heat transfer model was used to predict PCM's melting and solidification portions for every month. The result shows that the cooling effect is less in peak hot and peak cooled months conditions due to incomplete solidification and melting, respectively. The annual electrical output increased by 5.9 % in hot climate conditions. Mahamudul et al. [19] studied the effect of RT35 as PCM having 20 mm thickness integrated with PV panel under Malaysian climatic conditions, experimentally and numerically. The authors reported that panel temperature dropped by 10 °C and remained stable for 4–6 h. The PCM-filled rectangular container was integrated with a PV panel and its performance was tested at different inclination angles of the system [20,21]. The result shows that as the system inclination angle declines from 90° to 0°, the melting rate of PCM decreases, and panel temperature increases. This is due to the low convection current at lower inclination angles. The effect of weather parameters on the PV/PCM system was analyzed by Khanna et al. [22]. The result shows that the weather having less variation in ambient temperature increased electrical output by 9.7 %, while it dropped to 6.6 % with large variations. Also, the weather with higher solar insolation and low wind speed are favorable for PCM integration with the PV system. Shaito et al. [23] have evaluated the performance of PV panels integrated with four different PCMs in indoor conditions. Experiments were conducted on a 10 W PV panel with a constant heat flux of 1000 W/m^2. The result shows that calcium chloride hexahydrate as PCM has shown economic viability and electrical power enhancement compared to other PCM used.

2.2. PV + PCM rectangular enclosure + fins/metals and nanoparticles

The low thermal conductivity of PCM is a major concern among researchers when used in PV thermal management. The PCM having higher thermal conductivity is more suitable for PV applications was experimentally confirmed by Japs et al. [24]. Atkin et al. [25] have tested reference PV, PV-PCM infused graphite and PV-PCM-infused graphite + fin under simulated solar irradiance. The inclusion of graphite increases the PCM thermal conductivity from 0.25 W/mK to 16.6 W/mK. The PV-PCM-infused graphite + fin increase PV panel efficiency by 12.97 %. Insertion of fins into PCM is the viable solution to increase the thermal conductivity of PCM. The effect of fin interaction with PCM was numerically analyzed by Khanna et al. [26]. The authors reported that to keep PV panel temperature low, the distance between two fins should be 25 cm, fin thickness 2 mm and the length of the fin must touch to the container bottom. On the other hand, Klemm et al. [27] explored the incorporation of metal fiber into three different PCMs such as RT54HC, RT50 and RT44HC. The metal fiber porosity, PCM type, and PCM layer thickness were all investigated numerically using the COMSOL software. The result shows that RT54HC as PCM having 50 mm thickness with a high porosity results in a significant heat transfer and reduces PV panel temperature by 20 K. The thermal regulation of PV panels using PCM with foam thicknesses of 8 mm and 12 mm has been experimentally investigated in the environmental conditions of Taxila, Pakistan [28]. The result shows that a panel with PCM RT44 with 12 mm foam thickness maximally reduces panel temperature by 24.39 °C

Table 1
Properties of different phase change materials used in thermal management of PV panel.

PCM type	Density solid/liquid (kg/m^3)	Specific heat capacity (kJ/kg k)	Thermal conductivity (W/mk)	Latent heat of fusion (kJ/kg)	Melting temperature (°C)
RT42 [19]	880	2.0	0.2	145	38–43
White petroleum jelly [29]	900	2.5	0.18	N/A	36–60
RT18HC [35]	770/880	2.0	0.2	N/A	17–19
CL [36]	880/750	1.8/1.9	0.143	188	20.6
SP224A [36]	1490/1440	1.7	0.6	150	21.6
CP [36]	870/790	1.9/2.2	0.143	195	22.4
SP21E [35]	1400/1500	2.0	0.6	N/A	22–23
SP24E [35]	1400/1500	2.0	0.6	N/A	24–25
Pork fat [37]	910/896	2.03	0.36	170	36–45
PEG 1000 [38]	1092.7	N/A	N/A	159	38–40
Parrafin wax [39]	910/765	N/A	0.3	255	57
RT44 [28]	880/770	2	0.2	160	41–48
PT58 [28]	890/810	2.47	0.25	225	58

against a panel with only PCM of 20.95 °C and enhances electrical efficiency by 11.68 % and 12.06 %, respectively. Higher foam thickness results in higher PV panel performance compared to lower foam thickness.

In another study, Farouk Hachem [29] compared the performance of PV panels using pure PCM and combined PCM. The combined PCM includes white petroleum jelly as pure PCM mixed with copper and graphite powder. The result shows that combined PCM increases the average PV panel electrical efficiency by 5.8 % compared to pure PCM. Nada et al. [30] compared the performance of PV-PCM, and PV-PCM + Al_2O_3 nanoparticle (2 % by volume) with reference PV. PV-PCM system decrease temperature by 8.1 °C, with an increase in efficiency of 5.7 %. Whereas, PV-PCM + Al_2O_3 reduces PV panel temperature by 10.6 °C, which increases electrical efficiency by 13.2 %. A similar investigation was performed by H. E. Abdelrahman [31] using PCMs with Al_2O_3 nanoparticles and cylindrical fins. The result shows that the addition of Al_2O_3 to RT35HC PCM with cylindrical fins reduces the PV panel temperature by 52.3 %.

2.3. PV + other PCM encapsulations designs

A novel PCM encapsulation design integrated with PV panels was suggested by Akshayveer et al. [32]. The right wall profile design $\left\{y = (ax - b)^{\frac{1}{n}}\right\}$, where ($n$ = 1, 2 and 3) shows an increase in melting rate by 17 %, which results in a decrease in panel temperature by 11.5 %. As a result, electrical efficiency has increased by 12 %. Waqas et al.

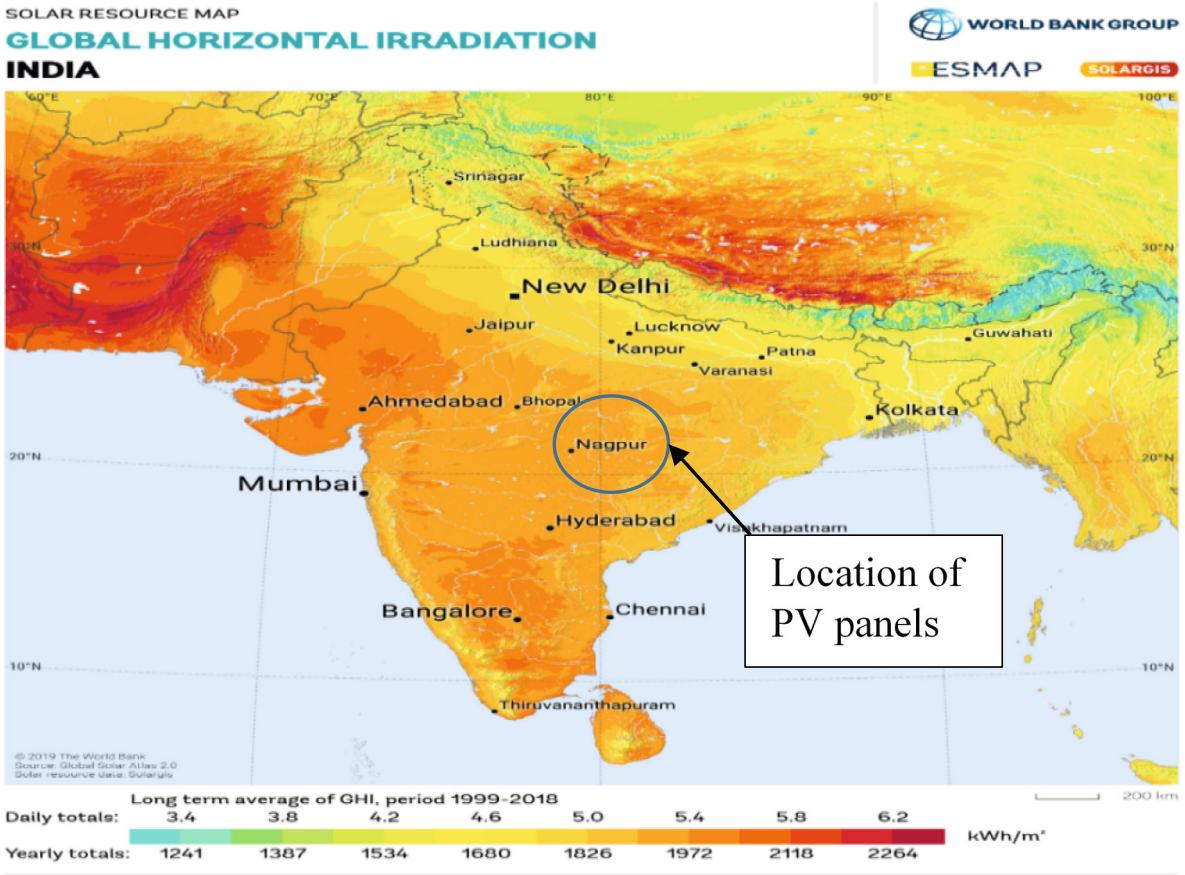


Fig. 2. Average daily/yearly horizontal global irradiation [37].

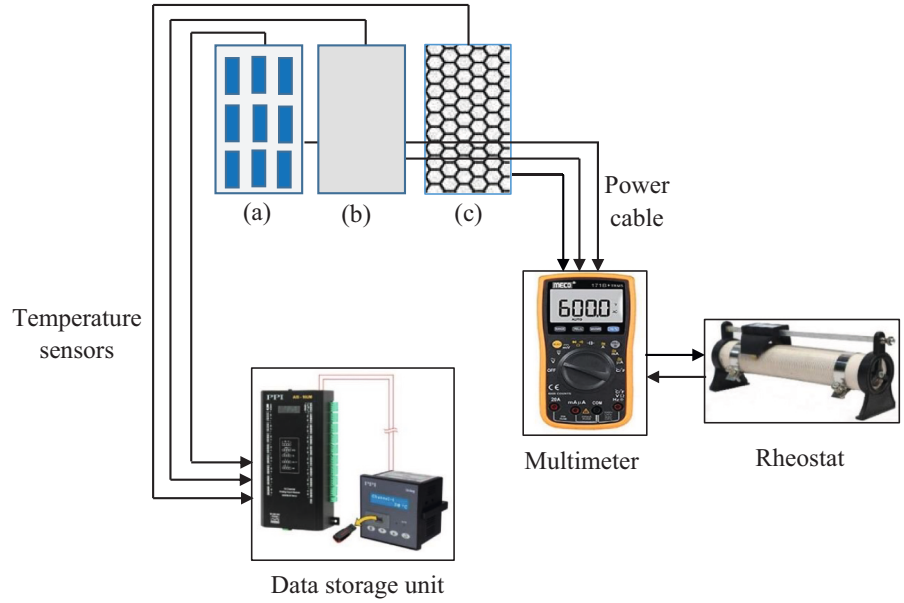


Fig. 3. Schematic illustration of the experimental setup: (a) reference panel; (b) PV-PCM and (c) PV-PCM with metal matrix systems.

[33] explored the effect of PCM-filled circular tube attached to the PV panel backside. These techniques reported a temperature drop of 8.5 °C compared to the reference panel. The use of PCM-filled circular tubes attached to the backside of the panel does not meet the requirement of uniform cooling. Emam et al. [34] showed a variety of heat sink arrangements for phase change materials attached to the panel backside. Compared to the single cavity, the three and five parallel cavity configuration shows lower PV panel temperatures.

Nowadays, numerous PCMs are used in the thermal management of PV panels. These PCM may be classified as organic, inorganic, hygroscopic, eutectic, and lastly, solid to solid PCMs. Table 1 shows the average characteristics of popular convectional and commercially available PCM materials for PV applications.

It is observed from Table 1 that different PCM materials have varying latent heat of fusion which typically range from 140 kJ/kg to 255 kJ/kg, and specific melting temperatures ranging from 20 °C to 60 °C. Another important property of PCM is the thermal conductivity which typically ranges from 0.15 W/mk to 0.6 W/mk. This property is most important in the system that features metal fins for more effective thermal management. The general selection of the specific type of PCM material is heavily influenced by the environmental conditions under which the PV-PCM system is intended to operate, as well as its economic feasibility.

In the present study, the use of Polyethylene Glycol 1500 as PCM in the thermal management of PV panels is discussed. It is a synthetic material (polyether compound) derived from petroleum. This polymer-grade PCM has a low cost, flammability, and toxicity advantages over paraffin. Therefore, the main objective of the present research is to study the effect of Polyethylene Glycol 1500 as phase change material on the thermal and electrical performance of PV panels in outdoor conditions of Nagpur, India. The average daily/yearly sum of global horizontal irradiation (GHI) available in India for power generation and other energy applications is shown in Fig. 2. From the figure, it can be seen that the study location receives 5–5.6 kWh/m^2/day on an inclined PV surface [40].

3. Experimental methodology

3.1. Experimental setup

Initially, four photovoltaic panels with thermocouple positions were tested for five days in outdoor conditions to know the variations in PV panel temperature. The experiments were conducted on three systems: PV (reference), PV-PCM and PV-PCM with metal matrix. The schematic of the experimental setup is shown in Fig. 3. The electrical properties of 40 W photovoltaic panels in polycrystalline nature at standard test conditions (STC) are listed in Table 2.

Table 2
Properties of PV Panels at STC (25 °C, 1000 W/m^2).

Cell type	Polycrystalline
Maximum power (Pmax) (W)	40 W
Maximum power voltage (V_{mp})	18.70 V
Maximum power current (I_{mp})	2.14 A
Open circuit voltage (V_{oc})	22.50 V
Short circuit current (I_{sc})	2.25 A
Module efficiency	13.48 %
Dimensions (mm)	670 (h) × 465 (w) × 30 (d)
Weight	3.3 kg

The PCM and metal matrix is packed on the backside of panel and covered with a transparent acrylic sheet. An epoxy solution is applied to make the system leakage free and fixing the thermocouple on the backside of PV panel. The fabricated experimental setup is shown in Fig. 4.

The honeycomb structure of the metal matrix having 6 mm thickness has been incorporated with PV-PCM to increase PCM thermal conductivity. The sample of a metal matrix having a honeycomb structure and thermocouple position on the backside PV panels can be seen in Fig. 5(a and b). Three thermocouples have been fixed on the backside of each PV panel. RTD PT 100 (3 wire and washer type) has been used to measure PV cell temperature. These RTDs are glued to the backside of PV panel with high-temperature adhesive tape (Kapton tape). A thermal imager (Testo 872, Germany) was used to capture the real-time thermal images of the PV panels during experiments. The detailed specifications of instruments/measuring devices used during experimentation are listed in Table 3.

The experimental setup is installed on the rooftop of the Mechanical Engineering Department of VNIT, Nagpur (21.1458° N, 79.0882° E), Maharashtra, India. Since India is positioned in the Northern hemisphere, the solar panel faces towards the south. As per experimental research done in Northern Hemisphere countries, 15° should be added/subtracted for the winter/summer season to the region's latitude to acquire the optimal estimated sun angle. Therefore, the total PV panel

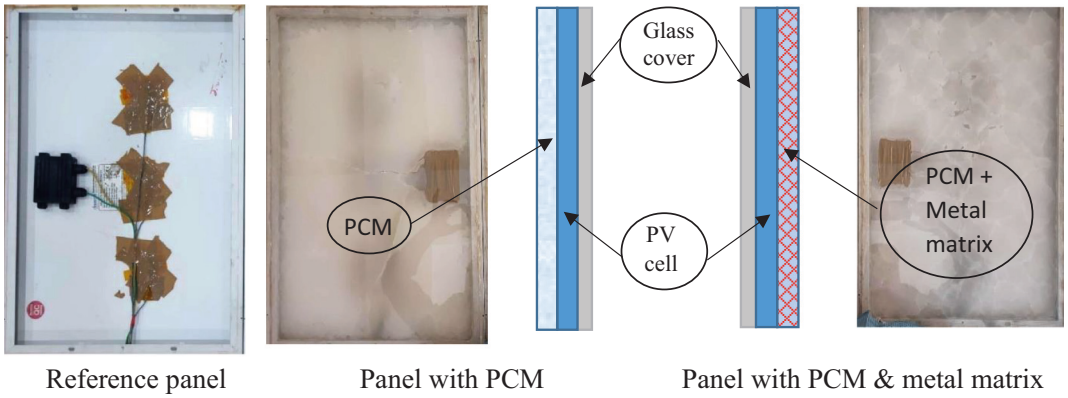


Fig. 4. Fabricated PV panels for experimental study.

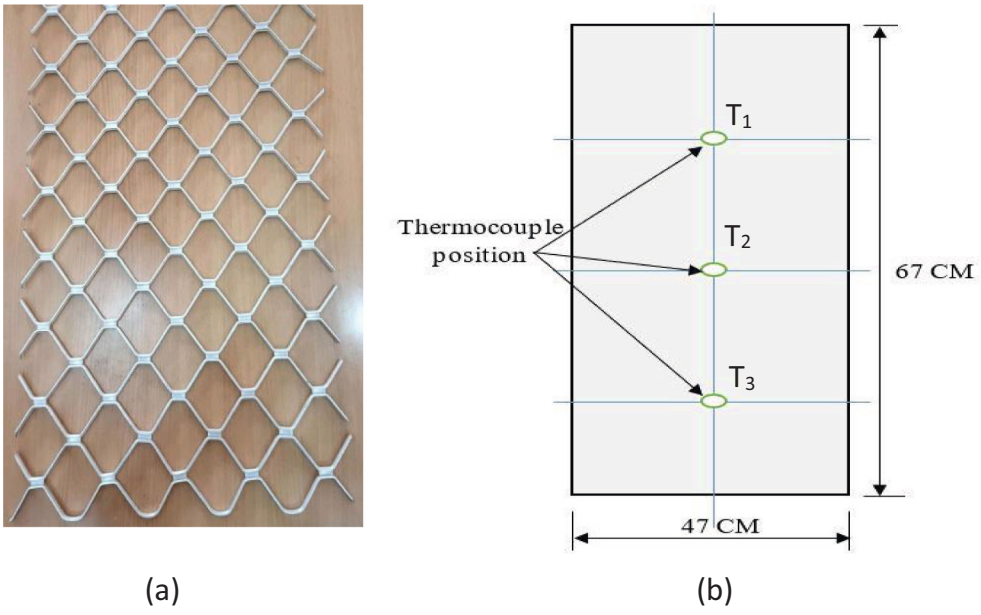


Fig. 5. a) Sample of honeycomb structure metal matrix; b) position of thermocouple on backside of PV panel.

inclination angle is equal to the latitude ±15°.

3.2. Phase change materials

According to the literature, one of the important thermophysical property of PCM is the melting point temperature. It plays a crucial role in controlling PV temperature in different climatic conditions. According to Shastri et al. [41], the melting point of PCM should be 10–12 °C higher than the ambient temperature of that site. In India, the summer ambient air temperature varies between 42 and 45 °C. Considering this condition, a Polyethylene Glycol (PEG) 1500 has been chosen as a phase change material. Differential Scanning Calorimetry (DSC) of PEG 1500 was carried out and graph obtained is shown in Fig. 6. During test the heating rate was 5 °C per minute. It is seen that the melting range of PEG 1500 is 50–56 °C. The properties of PEG 1500 are listed in Table 4. It can be seen that the properties of PEG 1500 does not deviate far from properties listed in Table 1. Finally, the melting point temperature range of PEG 1500 (50–56 °C) fulfills the climatic conditions of the chosen location.

3.3. Leakage proof system

The backside of PV panel is filled with PEG 1500 and it was covered with acrylic sheet using adhesive. During experiment, it was observed that PCM leakage from the system. As the temperature of PV panel increases, the melting of PCM start. This makes the system pressurized and PCM leakage occurs. To overcome this problem, a small hole is provided on the top side of the aluminum frame as shown in Fig. 7. After this correction, PCM leakage from system is almost stopped. This hole also helps in solidifying PCM at a faster rate.

3.4. Experimental procedure

The actual photograph of the experimental setup with measuring instruments in outdoor conditions is shown in Fig. 8. The experiments were performed for three consecutive days from 4th April 2022 to 6th April 2022 from 9:00 to 16:00 on clear sunny days. The panel's temperature was measured at 3 locations on the back surface of PV panel and recorded with the data storage unit. The thermal images (temperature distribution) of all PV panels have been captured with the help of thermal imager (Testo, model T-872) at maximum solar intensity. Climatic factors such as solar irradiance, ambient air temperature, and wind speed were also measured for the site.

Table 3
Detail specifications of the instruments/measurement devices used.

Sr. no.	Items	Model	Make	Descriptions
1.	Thermal Imager	T-872	Testo	Range: −30 to +100 °C; 0 to +650 °C Accuracy: ±2 °C, ±2 % of m.v.
2.	Pyranometer	Solar radiation recorder	Virtual electronics	Range: 0–2000 W/m^2 Calibration: 0.25 mV/W/m^2
3.	Temperature Sensor	RTD PT100	PPI	3 wire, washer type Range: −50 °C to 250 °C
4.	Multimeter	MAS830L	Mastech	Range: 0–10A DC Accuracy: standard: ±0.5 % of span
5.	Rheostat	DICO	TechDico	Range: 0–45 Ω, 3 A
6.	Anemometer	KM 908 MK1	MECO	Range: 0–45 m/s Accuracy: ±3 % Air temperature: 0–45°
7.	Data storage unit	Unilog	PPI	16-channel data recorder Accuracy: ±0.25 % of reading

3.5. Experimental uncertainty

To determine the PV power output and electrical efficiency, various measuring instruments were used. The measured parameters are subjected to uncertainties due to measuring errors associated with measuring instruments. Therefore, uncertainty analysis is carried out taking into account probable inaccuracies in voltage, current and radiation measurements. Parameters like voltage (V), current (I), and radiation (G) are measured and the possible errors in each are determined (see Table 5), which are then used to calculate the uncertainties related to experimental data of power output and efficiency of PV panel [43] using Eqs. (1) and (2):

$$\frac{U_p}{P} = \pm\frac{\sqrt{(VU_I)^2 + (IU_v)^2}}{P} \times 100 \tag{1}$$

$$\frac{U_\eta}{\eta} = \pm\frac{\sqrt{\left(\frac{UP_m}{AG}\right)^2 + \left(-\frac{P_m}{AG^2}U_G\right)^2 + \left(-\frac{P_m}{A^2G}U_A\right)}}{\eta} \times 100 \tag{2}$$

According to the calculations, the estimated uncertainty in the measurement of power output and efficiency are ±3.25 % and ±3.40 %, respectively.

4. Mathematical model

4.1. Energy balance of PV/PCM system

The energy balance equation is used in calculating the PV panel temperature rate using Eq. (3) (seen in Fig. 9)

$$\frac{dT_{PV}}{dt} = \frac{G - Q_{rad} - Q_{conv} - P}{C_{pv}} - \frac{Q_{cond}}{C_{pcm}} \tag{3}$$

where C_{pv} and C_{pcm} are the heat capacity of PV panel and PCM material.

The solar energy input (G) is given by:

$$G = \propto A\, I_r \tag{4}$$

where I_r — solar irradiance (W/m^2), $\propto$ — constant absorptivity and A is the panel area (m^2).

The heat transfer from PV to PCM material different modes of heat transfer play an important role.

a) The net radiative heat transfer from the system is given by:

$$Q_{rad} = 2h_r A\left(T_{pv} - T_\infty\right) \tag{5}$$

Table 4
Thermal properties of PEG 1500 [42].

Material property	PCM (PEG 1500)	Unit
Melting peak	53.1	°C
Density of solid, ρ_s	1120	kg/m^3
Density of liquid, ρ_l	1013.8	kg/m^3
Thermal conductivity, k	0.25	W/m-K
Latent heat of fusion, L_h	167–196	kJ/kg
Heat capacity, C_p	2100	J/kg-K

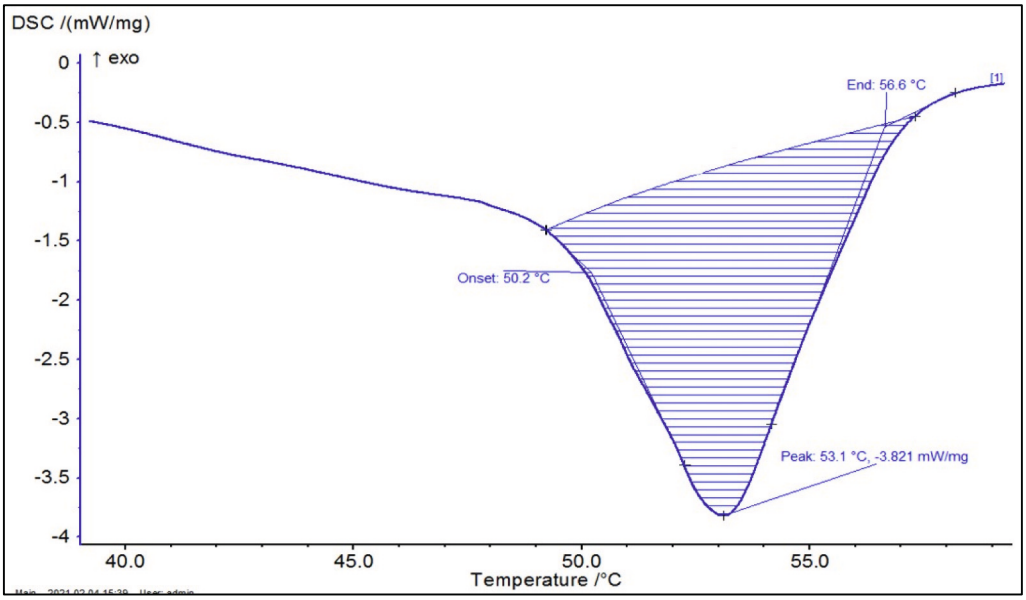


Fig. 6. Differential Scanning Calorimetry (DSC) heating curve of the PEG 1500.

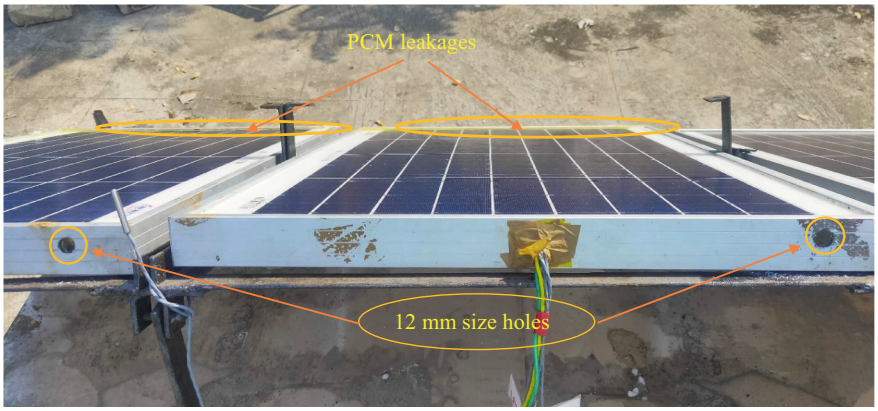


Fig. 7. Photograph of PV panel with a hole on top side.

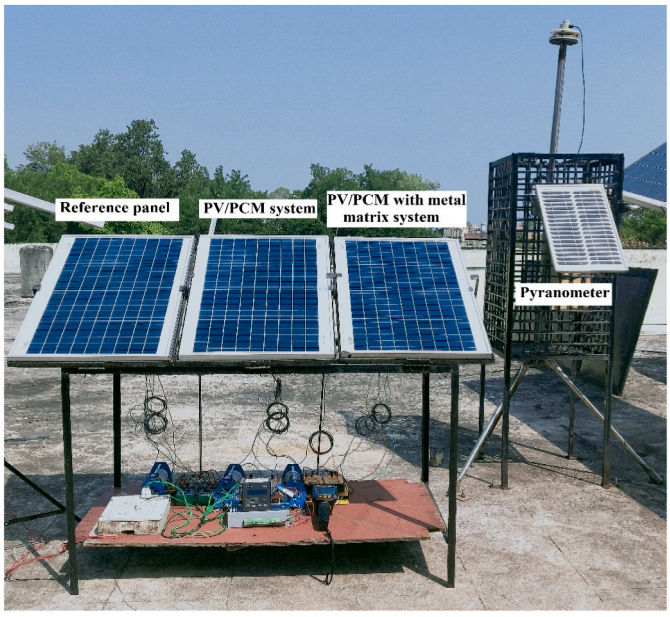


Fig. 8. Actual photograph of the experimental setup.

Table 5
uncertainty associated with measuring instruments.

Instruments	Standard error	Relative uncertainty
Thermocouple	1 °C	±3.87 %
Pyranometer	1 %	±1.76 %
Voltmeter	0.5 %	±5.27 %
Ammeter	0.5 %	±3.1 %

where T_∞ — ambient temperature and h_r — radiative heat transfer coefficient is given by Ramkumar et al. [44]:

$$h_r = \sigma\varepsilon\left(T^2_{pv} + T^2_{sky}\right)\left(T_{pv} + T_{sky}\right) \tag{6}$$

where σ — Stefan Boltzmann constant (5.67×10^{-8} W m^{-2} K^{-4}) and ε — emissivity of the panel. And where T_{sky} — sky temperature and it is given in modified form by Hendricks et al. [45];

$$T_{Sky} = 0.037536T_\infty^{1.5} + 0.32T_\infty \tag{7}$$

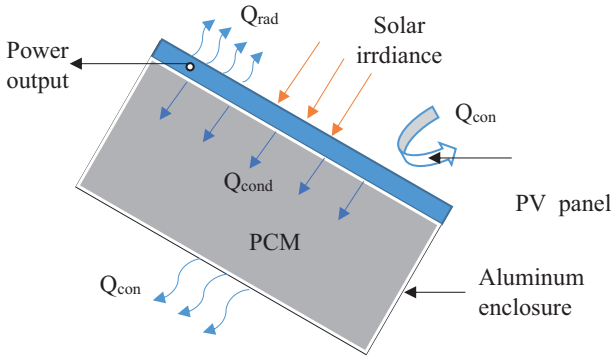


Fig. 9. Heat transfer model of the PV/PCM system.

b) The heat loss due to convection is defined as

$$Q_{conv} = h_c A(T_{PV} - T_\infty) \tag{8}$$

$$h_c = 5.7 + 3.8\,v \tag{9}$$

where h_c — heat loss coefficient and v — wind speed is given by Bayrak et al. [46];

c) Conduction heat transfer from PV panel to PCM is given by

$$Q_{cond} = Ak \times L_h(T_{PV} - T_\infty) \tag{10}$$

where x — thickness of PCM, k — thermal conductivity and L_h — latent heat of PCM material.

Hence, complete expression for change in temperature of the PV/PCM system from eq. 3 becomes;

The rate of change in temperature of PV panel

= Solar energy input − Different modes of heat transfer − power output

$$\frac{dT_{PV}}{dt} = \frac{\alpha\, A\, I_r - 2\, h_r A(T_{PV} - T_\infty) - h_c A(T_{PV} - T_\infty) - P}{C_{PV}} - \frac{Ak \times L_h(T_{PV} - T_\infty)}{C_{PCM}} \tag{11}$$

5. Result and discussions

The performance of PV/PCM and PV/PCM metal matrix systems are compared with reference PV system in outdoor conditions on clear three consecutive sunny days. The important performance parameters like PV temperature, temperature reduction, electrical efficiency and power output of the panels have been analyzed. In this paper, results obtained on 6th April 2022 are discussed.

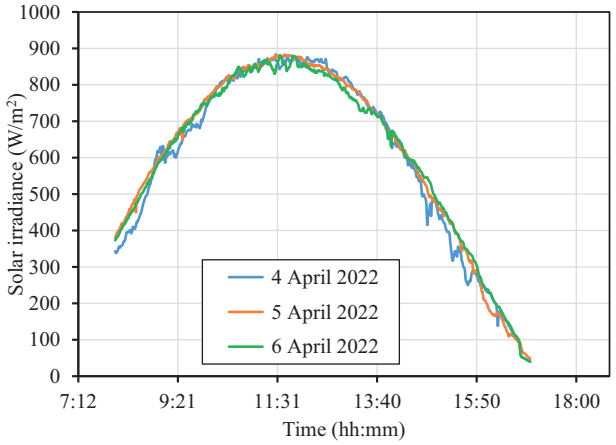


Fig. 10. Variations in the intensity of solar radiation with time for three consecutive days.

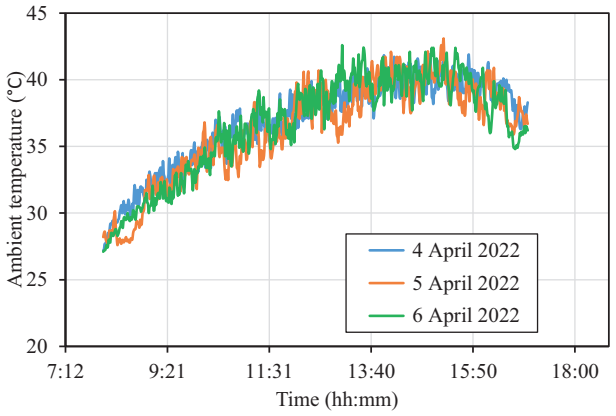


Fig. 11. Variations in the ambient temperature with time for three consecutive days.

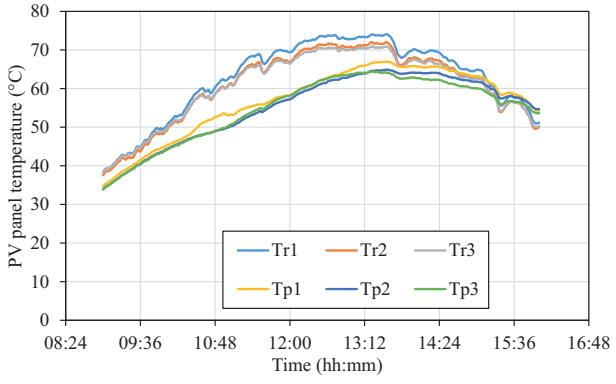


Fig. 12. Variations in temperatures of reference PV and PV/PCM system.

Fig. 10 illustrates the variations in the intensity of solar radiation with time for three consecutive days. It can be seen that in the early morning, solar intensity is less and when time progresses towards noon, the intensity is on the higher side (most solar energy reaches a solar collector) and then again falls at the end of the day. All three test days were totally clear and showed the same curve behavior throughout the day. The maximum solar intensity of 883 W/m^2 around 11:50 AM was recorded. Fig. 11 illustrates the little variations in ambient temperature. It can be seen that the highest air temperature was recorded between

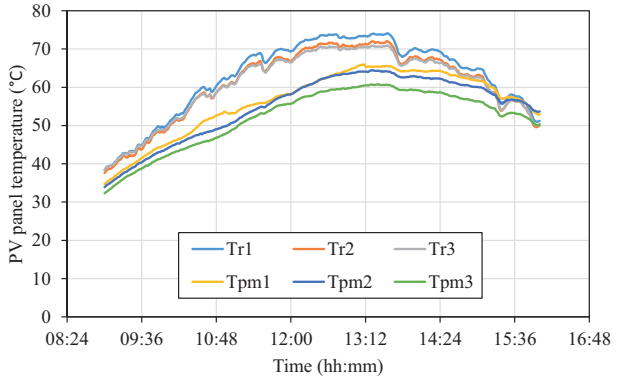


Fig. 13. Variations in temperatures of reference PV and PV/PCM metal matrix system.

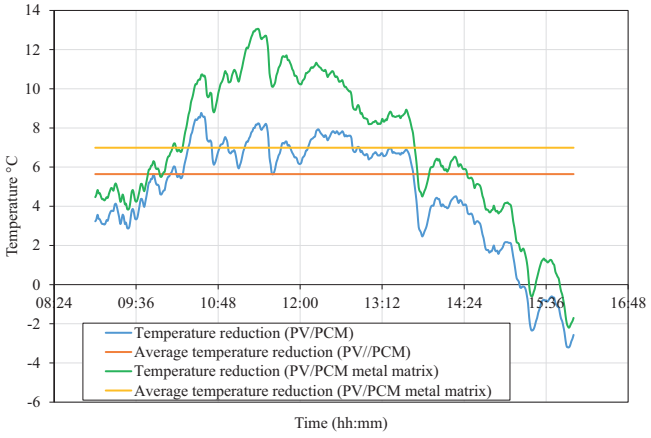


Fig. 14. Instantaneous and average temperature reduction with time for PV/PCM and PV/PCM metal matrix system.

12:00 and 15:00. Whereas the air temperature is usually low in the morning and end of the day. The maximum temperature recorded was 41.7 °C, 41.8 °C and 42.37 °C on the 4th, 5th and 6th of April 2022.

5.1. Temperature variations in PV panels

Solar irradiance has a significant influence on PV system performance. An increase in solar irradiance increases PV panel temperature. During experiments, the temperature across the panel at three different locations were recorded at an interval of 1 min. From 9:00 to 16:00 (on April 6, 2022) to identify the temperature distribution throughout the panel. Fig. 12 and Fig. 13 show variations in temperature for reference PV, PV/PCM system and PV/PCM metal matrix system, respectively. The Tr1, Tr2 and Tr3 curves denote the temperature of reference PV, whereas the Tp1, Tp2 and Tp3 curves denote the temperature of PV/PCM system (Fig. 12). Similarly, the Tpm1, Tpm2 and Tpm3 curves denote temperature of PV/PCM metal matrix system (Fig. 13). It can be seen that initially, all systems have nearly the same back surface temperature. However, as time progresses (solar radiation increases), reference PV temperature rises faster than cooled panels (due to presence of PCM). Addition of metal matrix in PCM improves the PV performance as it functions as a heat spreader. Thus, reduced temperature differential is formed inside the PCM layer due to homogeneous heat distribution. As a result, the rate of melting of PCM in PV/PCM metal matrix is lower than PV/PCM system. At 13:15, complete melting of PCM occurs and temperature of PV panels decreases. At 13:30, the highest temperature recorded for reference PV, PV/PCM and PV/PCM metal matrix are 72.33 °C, 65.4 °C and 63.4 °C, respectively. Also, it is

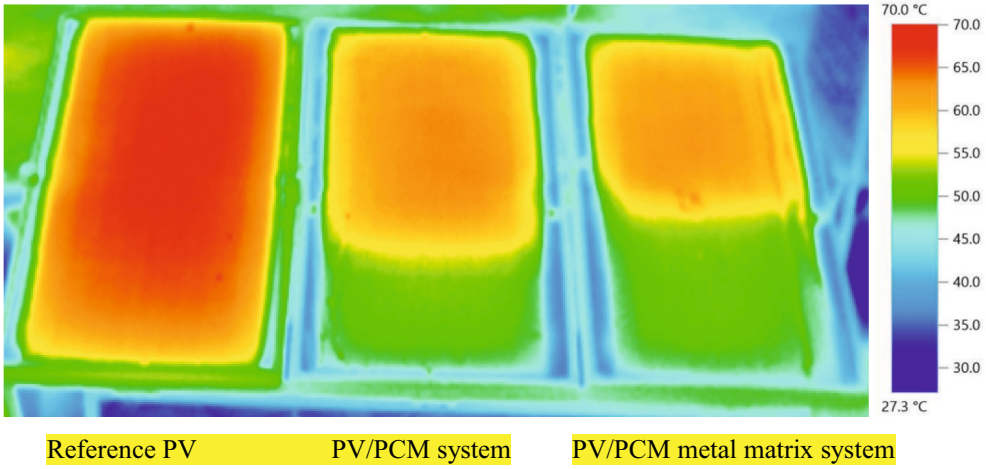


Fig. 15. Thermal images of the PV panels at maximum solar intensity (At 13:00, 6 April 2022).

observed that PV panel temperature can be maintained below 60 °C for 3 h in PV/PCM and PV/PCM metal matrix systems.

The temperature of all three panels were determined by taking the average of the temperatures recorded by each of the three thermocouples attached on the panel's backside. During the experiment, the average reference PV panel temperature (Tr1, Tr2, and Tr3) was 61.44 °C. Similarly, the average panel temperature of PV/PCM system (Tp1, Tp2 and Tp3) and PV/PCM metal matrix system (Tpm1, Tpm2 and Tpm3) were maintained at 56.72 °C and 54.44 °C, respectively. This shows an 8.32 % and 12.86 % temperature drop in panel temperature in PV/PCM and PV/PCM metal matrix systems, respectively.

The instantaneous and average temperature reduction across panels of reference PV, PV/PCM and PV/PCM metal matrix systems are shown in Fig. 14. From the figure, it can be seen that initially, at lower solar intensity, the average temperature drop in PV panels is small. As time progresses, the solar intensity increases and PCM starts melting by absorbing heat from panel and the temperature difference between reference PV and cooled panels increases. Once PCM in the container melts completely, the temperature difference between reference PV and cooled panels is almost negligible. Around 11:30, the maximum average temperature difference of about 9 °C and 13 °C was reported for PV/PCM and PV/PCM metal matrix systems, respectively. However, the average temperature difference over the whole day was 4.71 °C and 6.99 °C for PV/PCM and PV/PCM metal matrix systems, respectively.

5.2. Analysis of PV panels using thermal imager

Around 13:00, the temperatures of all three panels were captured using the thermal imager camera (Testo model 872) and the front surface temperature distribution obtained is illustrated in Fig. 15. It is recommended that such assessments should be carried out in bright sunlight with solar irradiation of at least 600 W/m^2 [47]. It can be seen that the surface temperature for the reference panel varies between 65 and 70 °C and the maximum temperature recorded with thermocouples attached to the backside of the panel was 72 °C. However, the thermal image captured shows a maximum of 70 °C. Around 2 °C higher temperature was recorded by thermocouples as they were directly in contact with the panel and measure more accurately than the thermal camera. Similarly, the surface temperature of the PV/PCM system has shown less temperature variation than the reference panel. The thermal image shows that around 60 % and 50 % surface area cover in the range of 55–62 °C for PV/PCM and PV/PCM metal matrix systems, respectively, on the top side. A PV/PCM metal matrix system shows around 10 % of the surface area has a lower temperature than PV/PCM system. Since the metal matrix acts as a heat spreader. These captured thermal images help to understand whether the cooling system employ uniformly cools the panel or not. Hence results in a higher electric power generation. Thus, the study of thermal images helps to analyze the effect of cooling system employed.

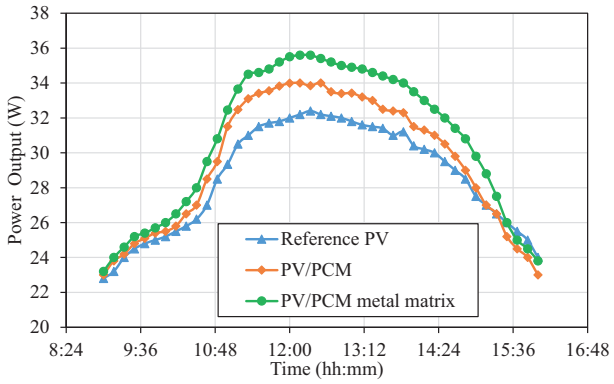


Fig. 16. Variations in the power output of PV panels with time (6 April 2022).

5.3. Power generation

The maximum power output (P_m) of the PV panel is defined as the product of the maximum voltage (V_m) and maximum current (I_m), obtained for the applied load and it is determined using Eq. (12).

$$P_m = V_m * I_m \quad (12)$$

Fig. 16 illustrates the variations in power output of reference PV, PV/PCM and PV/PCM metal matrix system with time. It can be seen that at the start of the experiments (in the morning) power output values are almost the same for all three panels. As time progress, power output of PV/PCM and PV/PCM metal matrix system increases compared to reference PV system. The maximum power output of reference PV, PV/PCM and PV/PCM metal matrix systems are 32 W, 34 W and 35.5 W, respectively. The PV/PCM metal matrix shows an additional 1.5 W increase in power output compared to PV/PCM system.

5.4. Electrical efficiency

The performance of PV panels is measured in terms of electrical efficiency. The efficiency of PV panel is strongly dependent on the PV cell temperature. The efficiency of PV panel is determined using the Eq. (13) suggested by Akshayveer et al. [32]:

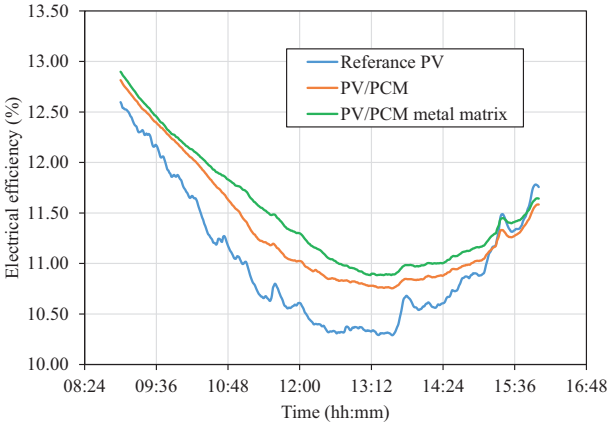


Fig. 17. Variations in the electrical efficiency of PV panels with time (6 April 2022).

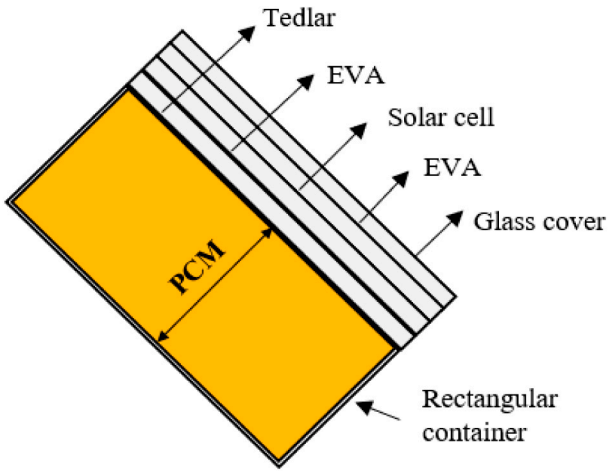


Fig. 18. PCM incorporation in a rectangular container.

$$\eta_{pv} = \eta_{ref}\left[1 - \beta_{ref}\left(T_{pv} - T_{ref}\right)\right] \quad (13)$$

where η_{pv} is the photovoltaic panel efficiency at PV cells temperature T_{pv}, η_{ref} is the reference efficiency at a reference temperature T_{ref} and β_{ref} is the temperature coefficient at a reference temperature T_{ref}. The values of η_{pv}, β_{ref} and T_{ref} specified by the manufacturer are 0.1348, 0.005/°C and 25 °C, respectively, which are close to the values specified for crystalline silicon panels given by Chandrasekar et al. [48]. The significant reduction in PV panel temperature using PCM and PCM metal matrix increases the electrical conversion efficiency. Fig. 17 illustrates the variations in electrical efficiency of reference panel, PV/PCM and PV/PCM metal matrix system during experiments. From the figure, it is clear that the cooling system had a significant effect on panel efficiency improvement. It can be seen that the electrical efficiency of PV/PCM and PV/PCM metal matrix systems shows higher values than the reference PV. During the period from 10:00 to 13:30, a maximum improvement in electrical efficiency has been observed. The average electrical efficiency of PV/PCM and PV/PCM metal matrix system over the day was 11.34 % and 11.52 %, against reference PV of 11 %, which shows an average increase in efficiency of 2.88 % and 4.24 % compared to reference PV. At 11:20 AM, the maximum electrical efficiency enhancement of 5.84 % and 8.87 % was reported for PV/PCM and PV/PCM metal matrix systems, respectively, compared to reference PV system.

6. Comparison between the current investigation and previously published studies

Several investigations has been performed on the thermal management of PV panel integrated with PCM [29,39,49–51]. In most of the investigations, a rectangular enclosure filled with PCM was utilized to completely cover the backside of the PV panel, as shown in Fig. 18. In the present study same approach has been used to incorporate the PCM in a rectangular enclosure. The results of the already published studies compared with the present study are presented in Table 6.

The previous studies selected for comparison with present study are based on the PCM melting temperature range with type of PCM used and cooling method employed. The present study demonstrates that PV panel temperature was significantly reduced with the cooling system used (PV/PCM metal matrix) which again improves thermal and electrical performance. The PCM (PEG 1500) used in the present study shows advantages over previously used PCM in cooling systems. Thus, use of PEG 1500 has wide scope in PV cooling systems.

7. Conclusions

A passive cooling system based on PV integration with PCM and metal matrix has been designed and developed. The Polyethylene Glycol 1500 is used as phase change material. An experimental study has been done for PV/PCM and PV/PCM metal matrix systems and compared with reference PV. The conclusions of the present study are as follows:

- The PV/PCM system significantly improved electrical efficiency and power output compared to reference PV. Addition of metal matrix in PCM (PV/PCM metal matrix system) shows enhancement in performance than PV/PCM system.
- The highest temperature recorded for reference PV, PV/PCM and PV/PCM metal matrix systems are 72.33 °C, 65.43 °C and 63.4 °C, respectively.
- With a significant decrease in PV panel temperature, a considerable increase in power output of 2 W and 3.5 W has been observed for PV/PCM and PV/PCM metal matrix systems, respectively, compared to reference PV.
- An average temperature drop of 8.32 % and 12.86 % over the whole day has been observed and a corresponding gain in electrical efficiency of 2.88 % and 4.24 % was reported for PV/PCM and PV/PCM metal matrix systems, respectively.

Table 6
Comparison between the results of the present study and previously published studies.

References	Cooling method	PCM used	Location	Temperature reduction	Electrical efficiency enhancement	Power output improvement
Ejaz et al. [27]	PCM + fiber	PT58 (58 °C)	Taxila, Pakistan	9.03 °C	5.9 %	11.77 %
Hachem et al. [29]	PCM + copper + graphite	White petroleum jelly (36–60 °C)	Lebanon	5.6 °C	5.8 %	7.4 %
Maiti et al. [39]	PCM + metal (V-trough)	Paraffin wax (56–58 °C)	Gujrat, India	16 °C	5.91 %	–
Present study	PCM + metal matrix	PEG 1500 (50–56 °C)	Nagpur, India	10–13 °C	8.87 %	10.93 %

- To increase overall efficiency of the system, the system can be combined with a water circulation network with phase change material.

CRediT authorship contribution statement

Someshwar S. Bhakre: Conceptualization, Investigation. **Pravin D. Sawarkar:** Writing – original draft, Writing – review & editing. **Vilas R. Kalamkar:** Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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